

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (20 points) Given a matrix $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$.

(a) Find the eigenvalues λ 's and corresponding eigenvectors x 's of A .

$$\begin{aligned} \text{Solution: } \det(A - \lambda I) &= \det \begin{bmatrix} 1-\lambda & 2 \\ 5 & 4-\lambda \end{bmatrix} \\ &= (1-\lambda)(4-\lambda) - 2 \cdot 5 \\ &= \lambda^2 - 5\lambda - 6 = (\lambda - 6)(\lambda + 1) \end{aligned}$$

$$(\lambda - 6)(\lambda + 1) = 0 \Rightarrow \lambda_1 = 6, \lambda_2 = -1$$

For $\lambda_1 = 6$, solve $(A - 6I)x_1 = 0$

$$\begin{bmatrix} -5 & 2 \\ 5 & -2 \end{bmatrix} x_1 = 0 \Rightarrow \begin{bmatrix} -5 & 2 \\ 0 & 0 \end{bmatrix} x_1 = 0 \Rightarrow x_1 = c \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

For $\lambda_2 = -1$, solve $(A + I)x_2 = 0$

$$\begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix} x_2 = 0 \Rightarrow \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} x_2 = 0 \Rightarrow x_2 = c \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

(b) Find an eigenvalue diagonalization $A = SAS^{-1}$ based on the results from (a).

$$\text{Solution: } S = \begin{bmatrix} 2 & 1 \\ 5 & -1 \end{bmatrix}, \quad S^{-1} = \frac{1}{-7} \begin{bmatrix} -1 & -1 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 1/7 & 1/7 \\ 5/7 & -2/7 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 2 & 1 \\ 5 & -1 \end{bmatrix} \underbrace{\begin{bmatrix} 6 & \\ & -1 \end{bmatrix}}_{\Lambda} \begin{bmatrix} 1/7 & 1/7 \\ 5/7 & -2/7 \end{bmatrix}$$